

On-line preconcentration of some rare earth elements in water samples using C₁₈-cartridge modified with 1-(2-pyridylazo) 2-naphthol (PAN) prior to simultaneous determination by inductively coupled plasma optical emission spectrometry (ICP–OES)

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Abstract

The on-line column preconcentration technique with inductively coupled plasma optical emission spectroscopy (ICP–OES) has been developed using a cartridge filled with octadecyl silica modified by 1-(2-pyridylazo) 2-naphthol (PAN). The aim of this method was to determine some rare earth elements (REEs) (Ce, Dy, La, Sm, and Y) and uranium in water samples. Sample solutions were passed through the C₁₈-modified column. The adsorbed cations were subsequently eluted from the column and transferred into the plasma with nitric acid solution for simultaneous determination of them. Sample pH, amount of PAN as a complexing agent, sampling and eluting flowrates and concentration of the eluent were optimized. Detection limits based on three times of standard deviations of blank by 10 replicates were in the range of 11 ng l^{−1} for Dy to 69 ng l^{−1} for U. Sample throughput was 10 samples h^{−1}. The proposed method was applied to determine REEs in natural water samples. Recoveries of the REEs from natural water samples were between 95 and 106% with percent relative standard deviation (%R.S.D.) of 1.0–7.9%.

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1. Introduction

In the last decades, the demand for rare earth elements (REEs) has increased because of their characteristics, such as the production of superconductors, super-magnets, and cathode-ray tube and geochemical nature. Since REEs are valuable indicators of petrogenic process, geologists and geochemists have investigated REEs in natural samples such as seawater [1–5], natural water [6], hydrothermal fluid [7], marine particulate [8] and rock reference materials [7–9], etc. The concentrations of REEs in rock reference material are in microgram per gram order [7–9], and those in the seawater

sample are still low level picogram per milliliter or less [1–6]. Recently, studies for determination of REEs in natural water samples have increased greatly [10–12].

Considering the low content of REEs in environmental samples, sensitive analytical techniques are required to obtain low limit of detections (LODs). Inductively coupled plasma optical emission spectrometry (ICP–OES) has become the most appropriate technique for their determination but the capability of this method is not sufficient when the concentrations of cations are too low. Also, the ICP–OES method sometimes suffers from problems, such as signal suppression and clogging of the sample introduction system when the sample contains dissolved salts at concentrations higher than 0.2% (w/v) [13,14]. On the other hand inductively coupled plasma mass spectrometry has been applied to determine REEs in the geological, natural water, and seawater samples. Although, ICP–MS has the detection power to de-

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termine trace elements at sub- $\mu\text{g l}^{-1}$ levels but it suffers from the problem of isobaric polyatomic interferences and also it is very expensive and its maintenance is very difficult [11].

To overcome these problems, many preconcentration techniques have been used including: co-precipitation with iron hydroxide [15–17], ion exchange with Chelex 100 [4,5,18,19], solid phase extraction (SPE) with silica-immobilized 8-hydroxyquinoline [6,20], solvent extraction with bis(2-ethylhexyl)hydrogen phosphate/2-ethylhexyl dihydrogen phosphate (HDEHP/H2MEHP) [21,22], and sorption on activated carbon [23]. It should be pointed out that co-precipitation with iron hydroxide requires an additional separation step, usually cation-exchange chromatography, to remove magnesium, calcium, and iron before determination. Chelation with Chelex 100 requires the removal of calcium and magnesium by careful washing with ammonium acetate or by cation and anion-exchange chromatography prior to elution of REEs [4,5]. The solvent extraction technique reported by Shabani et al. [21,22] involves many manipulations, such as scrubbing, stripping from the organic phase, washing the aqueous solution, and evaporating the final solution. The preconcentration technique reported by Esser et al. [6], and Halicz et al. [20], can only concentrate REEs at pH 8–9.

Moreover, batch preconcentration systems, having a risk of contamination, require relatively large amounts of sample and are time-consuming. However, on-line column preconcentration and elution systems can eliminate contaminations from various vessels and airborne particles in the overall procedure and require only a small amount of sample volume. As a result, more accurate and precise analytical values are easily obtained. Various adsorbent materials, such as Muromac A-I [11], iminodiacetate-based resin [12], 8-quinolinol-immobilized fluorinated metal alkoxide glass [MAF-8HQ] [10], poly(acrylamino)phosphonic dithiocarbamate) chelating fiber [1], Chelex 100 [4,9,6], activated alumina [24,25], activated carbon [23], HDEHP/H2MEHP adsorbed C_{18} cartridge [22], inorganic chemically active beads [26], aminocarboxylic sorbents [27], 1-phenyl-3-methyl-4-benzoylpyrazol-5-one coated on the inner walls of a PTFE mini column [28], nanometer-sized titanium dioxide [29] and octadecylsilica [30] were used for on-line preconcentration of REEs. For sorption of cations on an octadecylsilica column, the cation may be complexed in solution before the adsorption process takes place [30] or, by the ligand immobilized on a solid support.

1-(2-Pyridylazo) 2-naphthol (PAN) is a heterocyclic azo compound that has been used as a chelating agent for the SPE separation and preconcentration of metal ions [31,32].

The aim of this work was to develop a simple and sensitive method for the determination of some rare earth elements and uranium in aqueous matrices by ICP–OES using a C_{18} silica cartridge modified with PAN. The retention and elution conditions for simultaneous preconcentration and determination

by ICP–OES have been optimized. Finally, the developed method was verified by determination of these elements in various natural water samples and also seawater samples.

2. Experimental

2.1. Reagent

All chemicals were of analytical reagent grade and were obtained from Merck (Darmstadt, Germany). Deionized doubly distilled water was used throughout. All working standard solutions were prepared immediately before use, by stepwise dilutions of 1000 mg l^{-1} stock standard solutions (Alfa Aesar, Deutschland, Germany) to the required $\mu\text{g l}^{-1}$ levels. Acetate buffer solution (0.1 M) was prepared by dissolving sufficient amount of ammonium acetate in water and adjusting pH with 0.5 M nitric acid or 0.5 M sodium hydroxide solutions. The ligand solution (1400 mg l^{-1}) was prepared from dissolving analytical reagent grade PAN (Merck, Darmstadt, Germany) in methanol.

2.2. Apparatus

The measurements were performed with a simultaneous inductively coupled plasma optical emission spectrometry (ICP–OES, Varian Vista-Pro, Australia) coupled to a V-groove nebulizer and equipped with a charge coupled device (CCD). The ICP conditions are shown in Table 1. A two channel peristaltic pump was used for pumping the sample solution and eluent through the cartridge. The pH of solutions are determined and adjusted using a model 632 Metrohm pH meter with a combined glass-calomel electrode.

2.3. Procedure for on-line preconcentration

Schematic diagram of sample loading and elution of cations by on-line SPE and ICP determination is shown in Fig. 1. The on-line SPE and elution procedures are as follows: 15 ml of PAN solution (4.7 mg l^{-1} in distilled water) was pumped (by pump 1) at a flowrate of 10 ml min^{-1} through the SPE cartridge (Sep-Pak C_{18} cartridges, Part No. WAT051910, Waters Co., MA, USA) containing 360 mg C_{18} -silica sorbent with $50 \mu\text{m}$ particle size and $60 \text{ \AA}$ pore

Table 1
ICP–OES instrumental conditions

Plasma	Argon
RF generator power (kW)	1.5
Plasma gas flow rate (L min^{-1})	15
Auxiliary gas flow rate (L min^{-1})	1.5
Frequency of RF generator (MHz)	40
Observation high (mm)	8
Nebulizer pressure (kPa)	240
Solution uptake rate (ml min^{-1})	2

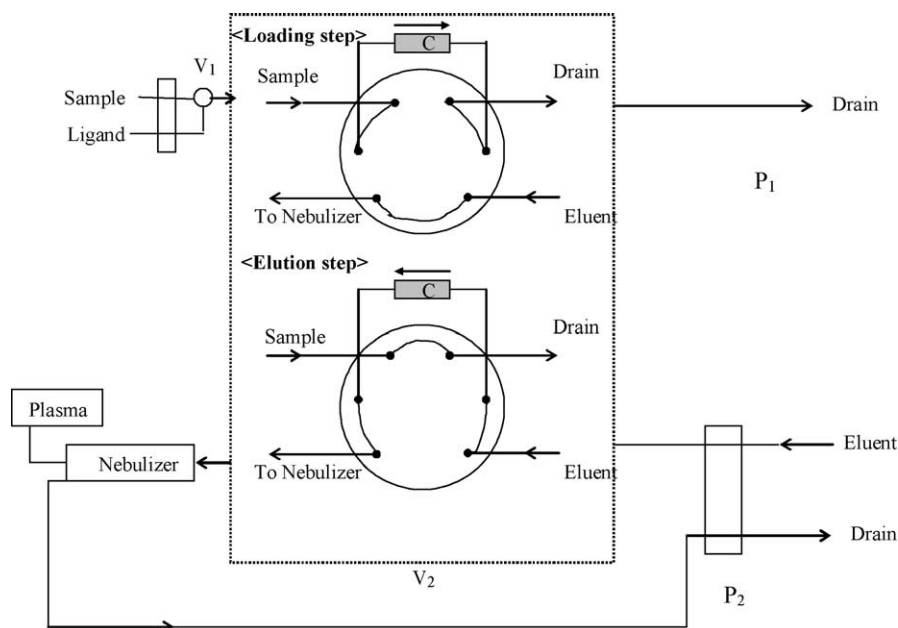


Fig. 1. Schematic diagram of on-line pre-concentration and simultaneous ICP–OES determination of uranium and REEs in water samples. P_1 and P_2 : peristaltic pumps, V_1 : three way valve, V_2 : injection valve, C: cartridge.

size. Then, 25 ml of sample solution was passed through the C_{18} -modified cartridge at a flow rate of 5 ml min^{-1} by pump 1 at loading valve position. Following sample loading, the cartridge was washed with 5 ml of double distilled water for 1 min to remove the remaining salts at the same flow rate. Then, the valve was switched to elution position; the REEs retained in the cartridge were eluted with nitric acid solution by pump 2. The eluted solution was directly introduced into the ICP nebulizer and the elution signals of elements were measured. On-line pre-concentration measurements were expressed as peak height emissions, which were corrected against the blank solutions. After that, the valve was switched to loading position again; the flow line and cartridge were washed with 5 ml methanol (for washing the adsorbed ligand) for 1 min and washed with 5 ml of double distilled water at the flow rate of 10 ml min^{-1} by pump 1. The column is now ready for the next sample introduction.

2.4. Method validation

In order to determine the validity of this method, the water samples were first filtered and then 100 ml of each was taken and its uranium and REEs contents were determined by proposed method. It was found that, in all samples, except the sea water, the amount of uranium and REEs was below the limit of detection of the method. Then, 150 ml of water sample was collected and divided into six portions of 25 ml. The proposed method was applied to three portions. Then, $1 \mu\text{g l}^{-1}$ quantities of REEs and uranium were added to the other aliquots of the sample and the elements were determined by the same method.

3. Results and discussion

No commercial ICP–OES instrumentation is capable for direct determination of the REEs in natural water samples due to their sub-ng l^{-1} concentrations in complex matrix [1,12,13,23]. Pre-concentration is an effective means of extending the detection limits of ICP–OES methods. This pre-concentration performed prior to the ICP–OES measurement permitted obtaining accurate and precise analytical results.

The optimized conditions for the enrichment procedure were established using 25 ml of REEs solution containing $100 \mu\text{g l}^{-1}$ of each element and submitting these solutions to the pre-concentration procedure. Influences of some parameters such as pH of the sample solution, amount of PAN, concentration of nitric acid as eluent, sample, and eluent flow rates, and interfering ions on the recovery of metal ions were studied.

3.1. Effect of the pH

One of the critical variables for the formation of REEs–PAN complex and its adsorption on the sorbent surface is the sample's pH. The effect of the sample pH on the REEs peak height was studied. The pH of solution prior to adsorption behaviors were adjusted in the range of 4.0–9.0 using NaOH and HNO_3 . Fig. 2 shows that the sorption of REEs on the modified C_{18} -silica did not occur under pH 4. Their sorption was increased with the raise of pH. The shape of the curve in Fig. 2 can be explained based on the equilibria: (1) At $\text{pHs} < 4$ the competition equilibria between cations and protons for occupying the ligand active sites occur, (2) At higher pH values ($\text{pH} > 8.5$) precipitation of the

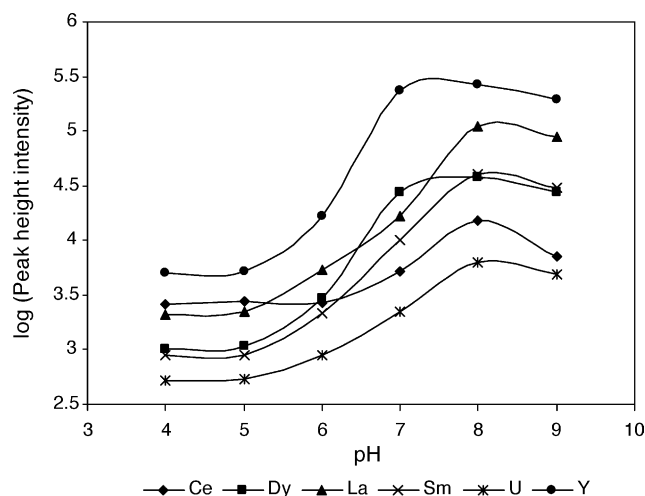


Fig. 2. Effect of pH of sample solution on the preconcentration of REEs.

salts occur. Considering these results, the selected pH was 8.0.

3.2. Effect of amount of the adsorbed complexing agent on C_{18} cartridge

The influence of the 1-(2-pyridylazo)-naphthol (PAN) amount was checked in the separation/preconcentration of Ce, Dy, La, Sm and U. The results showed that the emission signal of selected elements increased from 0 to 30 μg of PAN. The quantitatively extraction were obtained from 40 to 120 μg of ligand. On this basis, studies were carried out with 70 μg of PAN.

3.3. Effect of eluent concentration and flow rate

The elution of REEs from the cartridge was examined using nitric acid solutions of different concentrations (0.01–1.0 M) and the results are shown in Fig. 3. Total elution

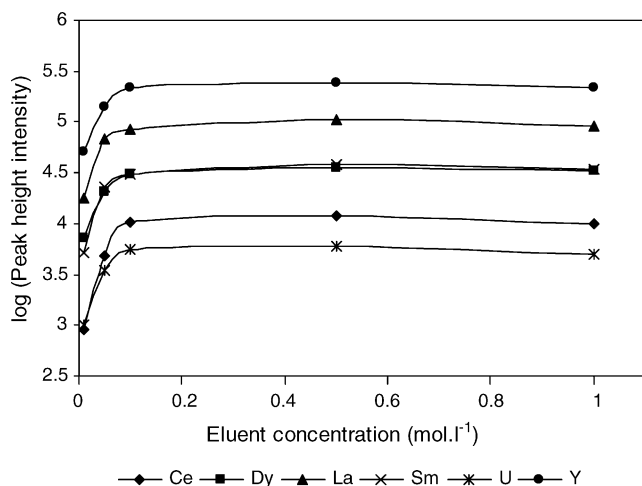


Fig. 3. Effect of eluent concentration on the elution efficiency of REEs from the C_{18} cartridge.

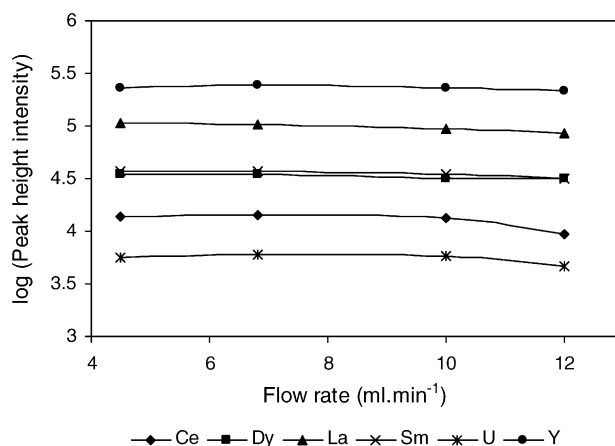


Fig. 4. Influence of sample loading flowrate on the peak height of REEs.

was reached at concentrations higher than 0.5 M. An increase in the acid concentration from 0.01 to 0.5 M results in narrowing the peak width and increasing its height. This effect becomes constant when the acid concentration is higher than 0.5 M and, therefore, this concentration of HNO_3 was chosen as optimum eluent concentration. The intensity of recorded signals must be sufficiently high and the peak widths should be narrow to obtain the maximum signal-to-noise ratio and minimum integration time. Therefore, the reversal of eluent flow through the cartridge during the elution step was applied. Also, the influence of the 0.5 M nitric acid solution flow rate in cations desorption from the cartridge was studied. The effect of eluent flowrate on the integrated intensity and peak height of elements signals during the elution step was investigated. Results showed that the analytical signal increased as the flowrate increased within the range of 1–3.4 ml min^{-1} due to decreasing the peak width. At flowrates higher than 3.4 ml min^{-1} analytical signal decreased due to incomplete elution of retained elements. Accordingly, 0.5 M nitric acid at flowrate of 3.4 ml min^{-1} was chosen for further studies.

3.4. Effect of sample flow rate

The flow rate of sample solution through the C_{18} -cartridge is a very important factor since this is one of the steps that controls the analysis time. The effect of the sample flowrate during the loading step on the emission intensities of REEs was studied in the range from 4.5 to 12.0 ml min^{-1} (Fig. 4). Results showed that the analytical signals did not vary significantly with the sample flow rate up to 12.0 ml min^{-1} . It means that the REEs sorption kinetics were very fast. Consequently, a flow rate of 12.0 ml min^{-1} was selected as an optimum sample flowrate.

3.5. Interference studies

The effect of the interfering ions on the recovery of REEs was studied with 10 $\mu\text{g l}^{-1}$ of REEs solutions prepared with and without each interference species. PAN does not form

Table 2
Effect of interferences on the determination of $10 \mu\text{g l}^{-1}$ Dy, Sm, U and Y in water samples

Interference	Concentration (mg l^{-1})	Recovery (%) ^a					
		Ce	Dy	La	Sm	U	Y
Ag^+	20 ^b	90 ^a	103	91	96	98	100
Hg^{2+}	40	95	90	92	98	94	93
Fe^{3+}	20	97	99	89	96	99	101
Cr^{6+}	16	91	94	91	97	101	98
Al^{3+}	15	103	99	101	98	103	94
Pb^{2+}	10 ^a	98	91	103	94	90	93
Mn^{2+}	18	99	103	99	100	101	100
Co^{2+}	60	93	97	94	95	103	100
Cu^{2+}	32	88	92	91	90	102	94
Zn^{2+}	42	91	97	88	96	100	96
Cd^{2+}	46	96	105	94	99	98	104
Ni^{2+}	53	87	92	95	89	99	93

^a Percent R.S.D.s varied in the range of 0.8–6.2% for three replicate measurements.

^b At higher concentrations precipitation was occurred.

stable complexes with alkaline and alkaline earth metal ions which are usually found in high concentration in natural waters. However, in order to identify their potential interference on the recovery of REEs ions on the modified C_{18} -cartridge, high concentration of them was tested. The alkali and alkali earth metal ions not affect the recovery of metal ions up to 1000-fold excess. Table 2 shows the tolerance limits of the interfering ions. The interfering species tested do not interfere with REEs retention at the studied concentrations. The obtained results indicate that the C_{18} -modified cartridge can be successfully used to extract REEs from high-saline water sample before their determination by ICP-OES. In addition, a number of common anions like Cl^- , Br^- , F^- , SO_4^{2-} and NO_3^- were tested and it was found that they did not interfere at concentrations up to 500 mg l^{-1} .

3.6. Analytical performance

In the analysis of real samples, the sample volume is one of the important parameters influencing the preconcentration factor. Therefore, the effect of the sample volume on the retention of Ce, Dy, La, Sm, U and Y was investigated. For this purpose, 50, 100, 150, 200, and 250 ml of sample solutions containing $10 \mu\text{g l}^{-1}$ of each cation were passed through the cartridge. The results indicate that the recoveries of the Dy, U and Y were close to 100% up to a sample volume of 250 ml,

but for Ce, La and Sm, this volume was up to 100, 100 and 150 ml, respectively.

The enhancement factors calculated as the ratio of the slopes of the calibration curves obtained by this flow method and by usual nebulization ICP-OES [33–36] were 328, 275, 337, 324, 382 and 372 for Ce, Dy, La, Sm, U and Y, respectively. It is worthy to note that in the present study for obtaining the above enhancement factors, the preconcentration times were 20 min for Dy, U and Y; 12 min for Ce and La and 8 min for Sm. Under the optimum conditions described above, the figure of merits of the method is described in Table 3. The reproducibility of the preconcentration method was evaluated by passing 25 ml of standard solution of REEs ($5 \mu\text{g l}^{-1}$) through the cartridge and repeating these procedure five times. The relative standard deviations (R.S.D.) were between 0.5 and 5.3% (Table 3). The detection limits (the detection limit was defined as the concentration that produces a signal equivalent to three times the standard deviation of 10 measurements of the blank) were 37, 11, 14, 23, 69, and 13 ng l^{-1} for Ce, Dy, La, Sm, U and Y, respectively. Dynamic linear range of the method was examined with the concentration of 0.5, 1, 5, 10, 100 and $200 \mu\text{g l}^{-1}$ REEs in water (the calculation of regression analysis was based on the peak height emission signals). The correlation coefficients of the calibration curves for REEs were in the range from 0.9994 to 0.9999. The overall time required for the on-line

Table 3
Figures of merit of the on-line preconcentration system for REEs determination

Element	Ce	Dy	La	Sm	U	Y
Enhancement factor	328	275	337	324	382	372
Calibration equation	$I = 301C + 57^a$	$I = 584C + 174$	$I = 1442C + 29$	$I = 557C + 120$	$I = 111C + 28$	$I = 4044C + 622$
Linear range ($\mu\text{g l}^{-1}$)	0.5–200	0.5–200	0.5–200	0.5–200	0.5–200	0.5–200
Correlation coefficient	0.9995	0.9997	0.9999	0.9998	0.9994	0.9999
%R.S.D. ^b	3.8	1.2	1.6	5.3	2.1	0.5
Detection limit (ng l^{-1}) ^c	37	11	14	23	69	13

^a The unit of C is $\mu\text{g l}^{-1}$.

^b Percent relative standard deviation for five replicate measurements of elements with concentration of $5 \mu\text{g l}^{-1}$ in water samples.

^c Detection limit based on $3S_d$ of blank signal.

extraction and ICP–OES determination based on ligand immobilization time (1.5 min at flow rate of 10 ml min^{-1}), SPE (2.5 min at a flow rate of 12 ml min^{-1}), washing of interferences (0.5 min at a flow rate of 10 ml min^{-1}), and elution of cations (1.5 min at flow rate of 3.4 ml min^{-1}) was about 6 min. Thus, the throughput was about $10 \text{ samples h}^{-1}$. Although the enhancement factor of the proposed method

is in the range of 275–384, the limit of detection of the method for REEs and uranium with respect to that of the ICP–OES cannot be reduced by this factor. This is due to the increased background signal in on-line-SPE–ICP–OES over that of the direct ICP–OES. Our methodology, in comparison with other on-line preconcentration methods that have been published previously, shows some similarities

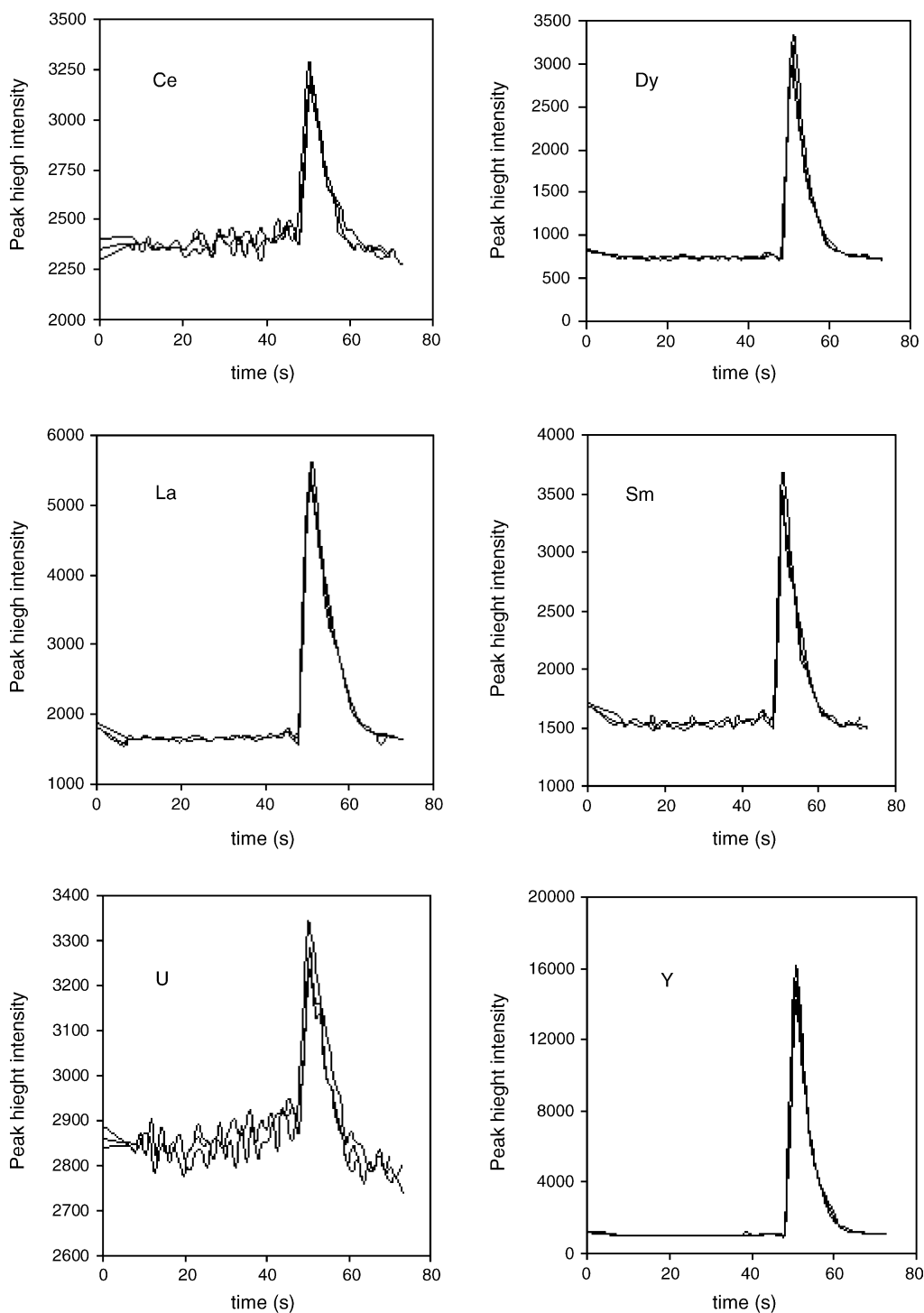


Fig. 5. The elution profile of REEs from the C₁₈ cartridge.

Table 4
Results of some REEs determination in natural water samples

Sample	Added ($\mu\text{g l}^{-1}$)	Found ($\mu\text{g l}^{-1}$) ^a						Recovery (%)					
		Ce	Dy	La	Sm	U	Y	Ce	Dy	La	Sm	U	Y
Spring	1.00	0.99 ± 0.06	0.97 ± 0.06	1.00 ± 0.04	1.04 ± 0.03	1.03 ± 0.05	0.99 ± 0.02	99	97	100	104	103	99
Well	1.00	0.97 ± 0.03	0.98 ± 0.08	0.98 ± 0.06	1.06 ± 0.02	1.05 ± 0.01	0.98 ± 0.05	97	98	99	106	105	98
Tap	1.00	0.98 ± 0.02	1.04 ± 0.04	1.02 ± 0.02	0.97 ± 0.01	1.06 ± 0.04	0.99 ± 0.08	98	104	102	97	106	99
Seawater ^b	1.00	0.99 ± 0.05	1.03 ± 0.03	1.00 ± 0.03	0.99 ± 0.04	7.22 ± 0.03	0.96 ± 0.02	99	103	100	99	95	96

^a Each result is the average of three replicate measurement with standard deviations of determinations.

^b The concentration of uranium in seawater which was obtained by Korkisch method [37] was $6.60 \pm 0.25 \mu\text{g l}^{-1}$. The concentration of other REEs in spring, well, tape and seawater were lower than LOD of the proposed method.

and differences. The detection limits that are obtained in this method are higher than those obtained using ICP–MS method [10–12,36]. However, it should be taken into account that the proposed ICP–OES method consists of much lower equipment and running costs. The precision of this method is similar to ICP–MS. The overall extraction time and sample volume in the proposed method are 6 min and 25 ml, respectively. The method proposed by Willie and Sturgeon [12] requires a 50 ml sample volume and 11 min for the extraction. The method proposed by Sohrin et al. [36] requires a 250 ml sample volume, 129 min for SPE extraction which is tedious and requires proficient techniques for pretreatment. Enhancement factors obtained in our method, ranged from 275 to 382, were higher than those obtained by Kajiya et al. [10], ranged from 7.8 to 16.4 ($t_{\text{loading}} = 2$ min; $F_{\text{sample}} = 3.2 \text{ ml min}^{-1}$); by Hirata et al. [11], ranged from 11.1 to 25.3 ($t_{\text{loading}} = 2$ min; $F_{\text{sample}} = 3.2 \text{ ml min}^{-1}$); by Zhang et al. [1], 200 ($V_{\text{sample}} = 1000 \text{ ml}$, $V_{\text{eluent}} = 5 \text{ ml}$, $F_{\text{sample}} = 7 \text{ ml min}^{-1}$); by Benkhedda et al. [28], ranged from 15 to 22 ($t_{\text{loading}} = 30 \text{ s}$, $F_{\text{sample}} = 4.4 \text{ ml min}^{-1}$). On the other hand enhancement factor of the present work is lower than obtained by Willie et al. [12], 500 ($V_{\text{sample}} = 50 \text{ ml}$, $V_{\text{eluent}} = 100 \mu\text{l}$, $F_{\text{sample}} = 5 \text{ ml min}^{-1}$). Also the enhancement factor of Willie et al. method is larger than our method but their flow injection manifold is complicated and sample throughput (5 h^{-1}) is lower than our method (10 h^{-1}). It is worthy to note that, although enhancement factors obtained in our work are larger than obtained in some other works but because of high sensitivity of ICP–MS in comparison with ICP–OES detection limits of the present work for REEs are higher than on-line-SPE–ICP–MS methods [1,10,11,28].

3.7. Application to natural and seawater samples

Natural water reference materials are not available with certified values for any of the REEs [14]. One of the major difficulties in quantification of these analytes in natural waters is their low concentrations ($<1 \text{ ng l}^{-1}$). Thus, the accuracy of the proposed method was tested by calculating the recovery of REEs from spiked tap, well, spring, and seawater samples. All the determinations were performed by aqueous standard solutions for calibration and without further matrix specific optimization. The peak heights of the transient signals were proportional to REEs concentration in the sample and were

used for all measurements. The recorded peaks for $1 \mu\text{g l}^{-1}$ seawater solutions were shown in Fig. 5. The recorded peaks are sharp and have a stable baseline. The certified values and the results are presented in Table 4. It can be seen that the percent recovery of REEs and uranium from water samples ranged from 95 to 106% and the performance of the method is very good in all types of natural waters. Finally, the relative standard deviations for REEs determination in the examined real water samples were located in the range of 1.0 and 7.9%. It is worthy to note that the uranium concentration in Caspian Sea is higher than those reported for other seas. It is surprising to say that one of the sources of nuclear pollution in the Caspian Sea is the Caspian Sea per se. Boris Gulubov reports [37]:

“In addition to man-made sources of radiation, the Caspian ecosystem collects and stores high levels of natural radioactive nuclides. Caspian waters, bottom sediments, and living organisms contain levels of uranium five to seven times higher than those in other seas, due mostly to the complex migration patterns of naturally occurring radioactive nuclides”.

To further investigate the accuracy of the proposed method, three different unspiked water samples with known uranium content (determined by the Korkisch method [38]) was obtained from Atomic Energy Organization of Iran and the amount of U(VI) in the 100 ml of samples was determined by the proposed method, as described in the Section 2. The resulting values are reported in Table 5. To test whether the resulting values with those obtained by the Korkisch method differ significantly we used paired *t*-test. Based on the paired

Table 5
Determination of U(VI) in different water samples

Sample	Korkisch method [37]	U(VI) concentration ($\mu\text{g l}^{-1}$) (proposed method)
Well number 1 (Anomali Bandarabas)	4.5 ± 0.2^a	5.18 ± 0.13^b
Well number 3 (Anomali Bandarabas)	8.1 ± 0.4	8.37 ± 0.1
Chadormelo water (Yazd, Iran)	16.8 ± 0.6	16.63 ± 0.27

^a Each result is the average of at least three replicate measurement with confidence intervals with confidence level of 95% (results reported by Atomic Energy Organization of Iran).

^b Each result is the average of three replicate measurement with standard deviations of determinations.

f -test the statistic “ t ” obtained from the following equation [39]:

$$t = dn^{1/2}/S_d \quad (1)$$

where d , S_d , and n are the mean, standard deviation of d , the difference between paired values and number of paired results respectively. Based on the results of Table 5, in the present work, $d = 0.26$, $S_d = 0.42$. Substituting in Eq. (1), with $n = 3$ gives $t = 1.06$. The critical value is $t_2 = 4.3$ ($P = 0.05$). Since the calculated value of t is less than 4.3, thus with 95% confidence, the present method gives significantly similar results for uranium concentrations in comparison with Korkisch method.

4. Conclusion

A rapid, simple and cheap method is proposed for simultaneous preconcentration and determination of Ce, Dy, La, Sm, Y and U in a variety of natural water samples. A low detection limit, up to 11 ng l^{-1} , an enhancement factor of 382 and a high sample throughput ($10 \text{ samples h}^{-1}$) are the main advantages of this analytical procedure in comparison with similar ICP–OES methods. The precision of the present method is similar to ICP–MS but detection limits for REEs are higher than on-line-SPE–ICP–MS methods. The proposed preconcentration procedure uses, for the first time PAN as complexing agent and sorption on C_{18} -silica in an on-line preconcentration system. This proposed method could be used for the preconcentration and determination of other REEs in natural water samples. Also, all preconcentration stages can be automatically controlled. The procedure is suitable for extraction of REEs from high matrix content like seawater and is an alternative method to ICP–MS for determination of uranium in seawater samples. The results of the present work showed that the uranium content of the Caspian Sea is higher than reported for other seawaters. In addition, the technique can be applied to the environmental samples without the risk of contamination from air particles and containers. It is interesting to note that, by combination of the on-line-SPE procedure proposed with ICP–MS the detection limit of the system can be lowered by a factor of about 100.

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